

Statistics II: Introduction to Inference

Topic 1

January 30, 2025

Population:

- Collection of all objects or measurements that are of interest.
- For example,
 - (i) average height of 18 years old boys in West Bengal in 2025,
 - (ii) expectation of a truncated $\mathcal{N}(0, 1)$ truncated on the region $[a = -2, b = 3]$.

Challenges/Constraints:

- Not practical:
 - (i) in terms of cost and time (for the first example), and
 - (ii) analytically intractable (for the second example).

Sample:

- A *subset* of the population.
- Should be a *good representative* of the population.

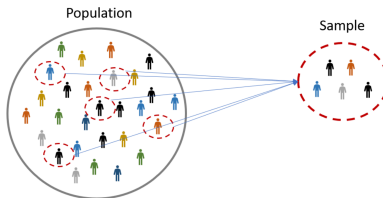


Figure 1: Illustration of population and sample.

What is a Good Representative?

Good Sample

A random sample of 18 years old boys from all over West Bengal.

- accurately reflects the population.
- randomly selected to avoid bias.

Bad Sample

A random sample of 18 years old boys only enrolled in the BSDS course.

- fails to represent the population.
- selected with bias or convenience.

Drawing a Sample (Simulation)

We draw random samples $\{X_1, X_2, \dots, X_n\}$ from the population.

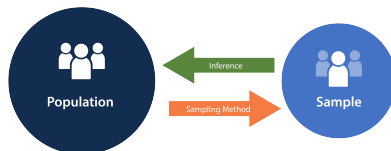


Figure 2: Sampling from the population to infer its characteristics.

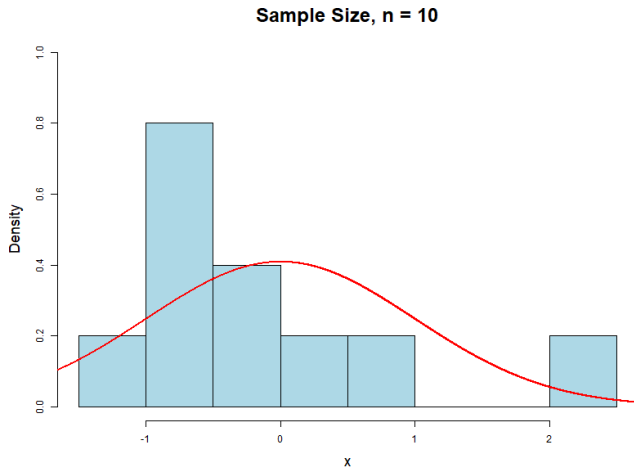
Here is an example of R code:

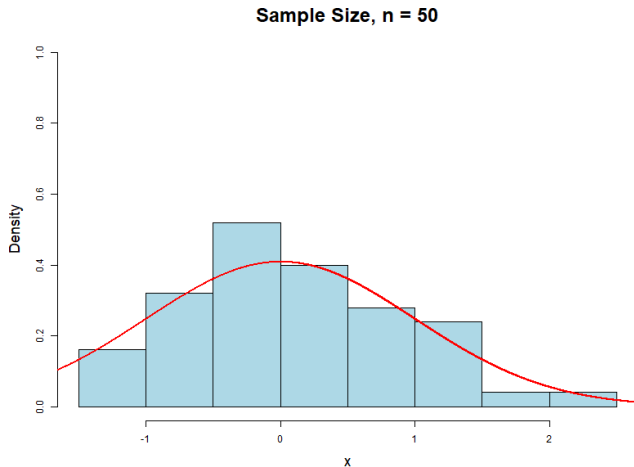
```
1 # Draw 1000 random samples from a  $N(0,1)$ 
2 x = rnorm(n = 1000, mean = 0, sd = 1)
3 # Draw 1000 random samples from a Poisson ( $\lambda = 2$ )
4 x = rpois(n = 1000, lambda = 2)
```

Why is Sample Important?

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample from the population variable X (say), which follows truncated $\mathcal{N}(0, 1)$ with $a = -2, b = 3$.

Observe how histograms are changing with increasing the sample size (n) for example (ii).





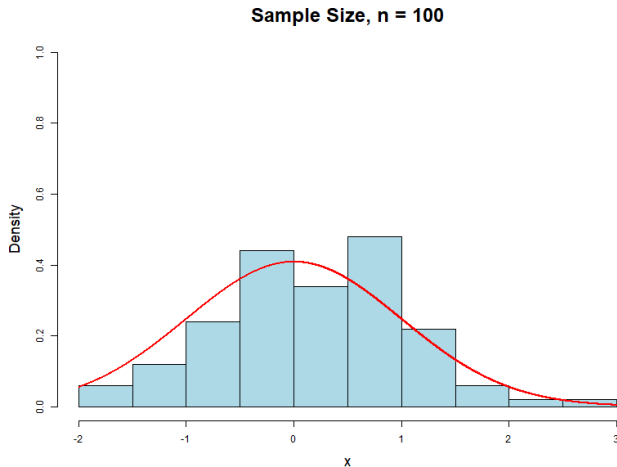
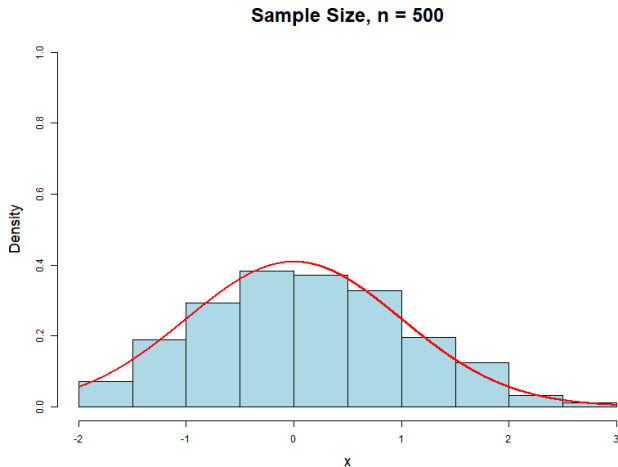
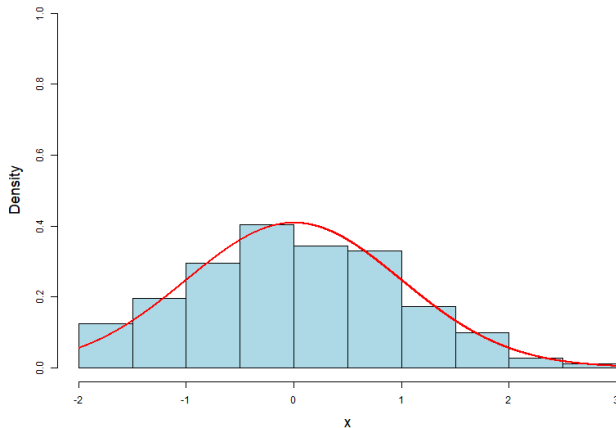


Figure 3: Histogram-3



Sample Size, $n = 1000$



Model: For example (i),

- Assumption 1: X is *continuous*, where X denotes the height of a randomly selected 18 years old boy.
- Assumption 2: $X_1, X_2, \dots, X_n$ are independently and identically distributed (*i.i.d.*) samples drawn from the distribution of X .

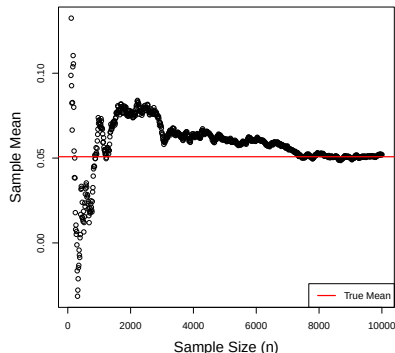
Parameter:

- $\theta = \mathbb{E}(X)$: the average height of the entire population.
- $p = \mathbb{P}(X \in [a, b])$: the probability of those 18 years old boys whose heights are within $[a, b]$.

Statistical Regularity

Let $X_1, X_2, \dots, X_n$ are *i.i.d.* copies of the random variable X as defined in example (ii), then the parameter of interest may be $\mu = \mathbb{E}(X)$ and $\bar{X}_n$ is the sample mean. The statistical regularity says

$$\bar{X}_n \rightarrow \mu \text{ as } n \rightarrow \infty.$$



R Code Example

Here is an example of R code:

```
1 library(truncnorm)
2 true.mean = etruncnorm(a = -2, b = 3)
3 n = seq(100, 10000, by = 10)
4 x = rtruncnorm(max(n), a = -2, b = 3)
5 x.bar = c(0)
6 for (i in 1:length(n)) {
7   x.bar[i] = mean(x[1:n[i]])
8 }
9 plot(n, x.bar, cex.lab = 1.5)
10 abline(h = true.mean, col = "red")
11 legend("bottomright", legend = c("True Mean"), col = c("red"))
```

- 1 Consider a random sample with size $n = 5$ from an exponential distribution with rate parameter $\alpha = 3$. Check how the histograms are changing with increasing the value of n .
- 2 Consider a sample of size n drawn from a Poisson ($\lambda = 5$) distribution. Does the sample mean converge to the population mean if we increase the sample size? (start with $n = 10$)
- 3 Suppose $X_1, \dots, X_n \sim \text{Bernoulli}$ distribution with parameter $p \in (0, 1)$. Prove that $\frac{1}{n} \sum_{i=1}^n X_i^2$ converges to p if the sample size increases. (take $p = 0.67$ and start with $n = 8$)